

Homework Assignment 7

Bhabha Scattering

All problems are graded on effort only.

In this problem set, we will consider the QED process

$$e^- + e^+ \rightarrow e^- + e^+$$

which is known as Bhabha scattering. We will label the particles as:

$$a + b \rightarrow 1 + 2$$

so e.g. the final state e^+ is particle 2. We will consider the limit that $m_e \ll E$. In this limit, the helicity eigenstates for particles are given by:

$$u^\uparrow = \sqrt{E} \begin{pmatrix} c \\ se^{i\phi} \\ c \\ se^{i\phi} \end{pmatrix}, \quad u^\downarrow = \sqrt{E} \begin{pmatrix} -s \\ ce^{i\phi} \\ s \\ -ce^{i\phi} \end{pmatrix},$$

and the helicity eigenstates for anti-particles are given by:

$$v^\uparrow = \sqrt{E} \begin{pmatrix} s \\ -ce^{i\phi} \\ -s \\ ce^{i\phi} \end{pmatrix}, \quad v^\downarrow = \sqrt{E} \begin{pmatrix} c \\ se^{i\phi} \\ c \\ se^{i\phi} \end{pmatrix},$$

where in both cases:

$$c \equiv \cos\left(\frac{\theta}{2}\right), \quad s \equiv \sin\left(\frac{\theta}{2}\right).$$

Problem 1: Show that the Matrix element for this process is:

$$M = -\frac{g_e^2}{s}(j_{ab})^\mu(j_{12})^\mu - \frac{g_e^2}{t}(j_{a1})^\mu(j_{b2})^\mu$$

for suitably defined currents j_{ab} , j_{12} , j_{a1} , j_{b2} and with s and t the Mandelstam variables. Use a shorthand like $u(a)$, $v(2)$ for the bispinors and notice that the helicity states have not been specified yet.

Problem 2: Why is there a contribution from both s and t channel, but no contribution from u channel?

Problem 3: Show that:

$$M_{RL \rightarrow RL} = -g_e^2(1 + \cos \theta) \left(1 - \frac{s}{t}\right) = -g_e^2(1 + \cos \theta) \left(1 + \frac{2}{1 - \cos \theta}\right)$$

where θ is the angle between the initial and final state electrons.

Problem 4: Use Fermi's golden rule to calculate the scattering cross-section

$$\frac{d\sigma}{d\Omega}$$

for this process in the CMS with helicity $RL \rightarrow RL$.